Inbrev

Hack Apertus 2026 · Track 2B · Team Apertool

Run locally, the practice inbox never leaves the practice.

Inbrev sorts the inbox of a Swiss GP practice by handling path and deadline, with Apertus, built to run on a box in the practice. Every field it extracts quotes its source.

▶ Watch the 2-min demo

Code on GitHub

solo builder · fictional data only

THE PROBLEM

Before the first patient, there is the inbox.

Lab reports, refills, insurer forms and faxes in four languages land in one queue. A practice assistant (MPA) reads them one by one.

9.5 h / week

of administration for Swiss GPs, up from 8.6 h in 2020

Workforce Study, *Praxis* 2026 · n = 1,776

54 min / day

on requests from authorities and insurers alone

gfs.bern for FMH, 2025 · N = 1,532

Art. 321

Professional secrecy binds doctors **and their assistants**

Swiss Criminal Code · FADP Art. 5

HOW IT WORKS · PURPOSEFUL USE OF AI

Sorted by who handles it and by when.

01 · READ

Mail, forms, faxes

E-mail, IMAP, web forms, fax via OCR. German, French, Italian, English.

02 · ROUTE

9 handling paths

Plus the administrative deadline and who takes it.

03 · PROVE

Quote or drop

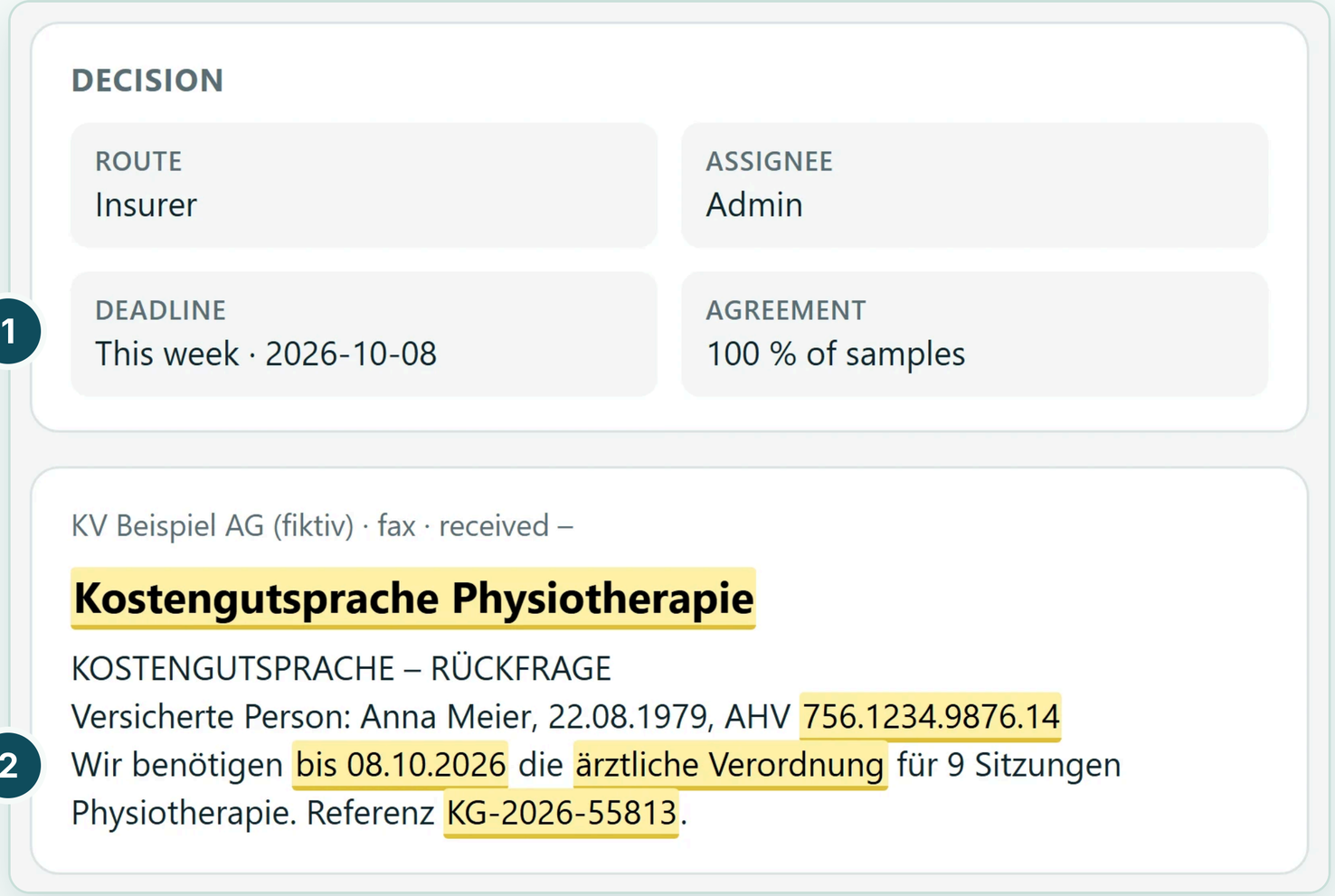
Every field quotes its source. No quote in the original, no field.

Clinical content Possible symptoms leave automation and go to a person. Not rated, not ranked.

Code checks what code can check: AHV numbers, dates, quotes.

DEMO

One fax, sorted. Every value shows where it came from.



1 **Route and deadline**

Insurer, admin, by 8 Oct. The date comes from the quote, the deadline from code.

2 **Quote or drop**

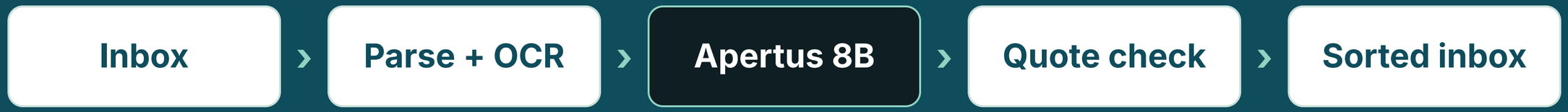
Each highlighted value is verbatim in the fax. No quote, no value.

Running app, stored Apertus 8B result, fictional fax. Possible symptoms would go to a person instead.

ARCHITECTURE · SOVEREIGN DEPLOYABILITY

Pull the cable. Inbrev keeps sorting.

INSIDE THE PRACTICE · NO OUTBOUND CONNECTION



On-prem	mini PC in the practice
Air-gapped	mail gateway hands messages in
Swiss cloud	option for single practices

```
Docker, no network
egress dns / tcp / https: FAILED
Local Q4, laptop iGPU
71.8 s / message, 16/16 clinical out
```

EVALUATION · TECHNICAL RIGOUR

No synthetic clinical message ran by itself.

180 synthetic messages, labels in human check (results on 16 October).

Setup	Routes right	Clinical out	Runs by itself
Keyword rules + lexical net	0.79	7/16	88 %
Apertus 8B, one answer, no checks	0.87	9/16	95 %
Inbrev with Apertus 70B	0.96	15/16	82 %
Inbrev 8B + symptom pass	**0.94**	**16/16**	**78 %**

Paired on 180: 8B beats the rules 32 to 5 ($p = 7 \cdot 10^{-6}$). Details and all rows: report §5.

16/16

clinical messages out of automation
95 % CI 0.81–1.00 (Wilson)

- Labels fixed before any text; Apertus never writes or labels test data.
- Two open models (not people) relabelled 367 messages blind: κ 0.93 route, 0.97–0.98 clinical. Human check in progress – results on 16 October.

PURPOSEFUL USE OF AI · MODEL CHOICE

Why Apertus, and what it costs.

Measured, same pipeline

- ✕ Qwen3.5-9B, Ministral 8B: 177/180 routes
- ✕ Apertus 8B: 170/180 · 70B: 172/180
- ✕ Fresh hold-out: 5 wrong routes run by themselves (Qwen 0); 70B second opinion: 3

Why Apertus anyway

- ✓ Only openly available training data, opt-outs applied, personal data filtered
- ✓ Published data catalogue and EU training-content summary
- ✓ Local run and quote checks work with any model: no argument

A trade-off, not a free choice. Sources and hold-out details: report §5.

VALUE, COST & SCALABILITY

Swiss francs per 1,000 messages, on hardware the practice owns.

ON-PREM MINI PC

≈ CHF 12

at 60 messages a day; box, power, CPU time before the symptom pass

SWISS CLOUD VM

CHF 2.39

billed while sorting; same CPU time, other CPU not measured

SWISS TOKEN API

CHF 3.13

Apertus 70B list price, tokens measured with the symptom pass

≈ 32 min

MPA time saved a day ASSUMPTION

60 messages × 78 % automatic × 45 s saved, minus rework of misroutes

Estimates, prices excl. VAT, 4 Oct 2026. Box and value: report §5; VM: `scripts/cost_model.py`.

INTENDED PURPOSE · IMPLEMENTATION FEASIBILITY

Administrative by default. Clinical mode is research only.

Default mode: Inbrev does

- ✓ Sort by path and deadline
- ✓ Extract fields, each with its quote
- ✓ Say “you decide” when unsure

Default mode: Inbrev never

- ✕ Rank by clinical severity
- ✕ Rate symptom severity or suggest care
- ✕ Send patient data out (local mode)

Clinical research mode: off by default, not validated, not certified (MDCG 2019-11, MepV Art. 15).

BUSINESS MODEL & ROADMAP

A module for the software practices already use.

User	The MPA, first thing in the morning
Buyer	Practice owner or doctors' network
Price	CHF 150–250 / month HYPOTHESIS
Channel	Module in HIN mail or practice software

OCT 2026
Evaluation, offline proof, cost

NOV – DEC 2026
5–10 interviews with MPAs and owners

Q1 2027
Pilot in one group practice

Q2 2027
Classification review before first sale

Inbrev

Team Apertool · Niklas Wichter, solo builder

OUR ASK

Pilot Inbrev in one Swiss group practice.

On its own box, from Q1 2027. Your inbox stays in your practice.

Contact: hello@apertool.ch

inbrev.apertool.ch

Demo ▶ 2-min video

Code github.com/NWichter/Hack_Apertus_2B_Inbrev